

Group Assignment 2: Pilgrim Bank A & B

Group 8: Team Deadline Daredevils

Aniruddh Maniktala (37332889)

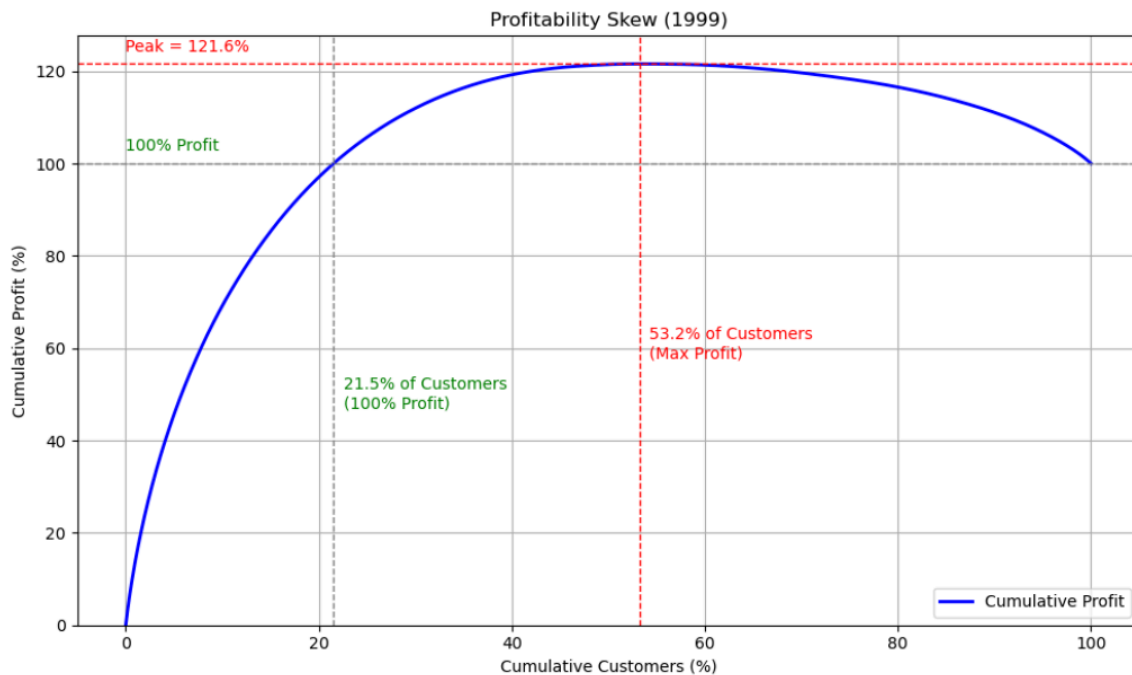
Chion Kim (87525523)

Reshum Zubair (91807925)

Saloni Gharge (55859177)

Question 1:

Re-create the “profitability skew” discussed in the case (see Exhibit 3).



Question 2:

Evaluate the heterogeneity in customer profitability:

a. Currently Pilgrim Bank has 31,634 customers that generate 100% of profit. They could realize the same profit with less customers. Determine the minimum percentage of customers that Pilgrim Bank would need to realize 100% of their current profit.

Ans - Minimum percentage of customers needed to achieve 100% of profit- 21.53%

Pilgrim Bank currently has **31,634** customers, but only **21.53%** of them (approximately **6,811 customers**) generate the full **100%** of the bank's profit. This highlights that the majority of customers contribute little or even negatively to profitability.

b. Determine the percentage of total profit Pilgrim Bank could realize by targeting profitable customers only. What is the percentage of customers they need to realize this maximum profit?

Ans- Targeting only profitable customers

By focusing solely on profitable customers (those with positive profit), Pilgrim Bank could achieve **121.58%** of its total profit. This suggests that unprofitable customers significantly drag down overall profitability. These **profitable customers make up 53.21%** of the total customer base, implying that targeting these customers could enhance profitability while reducing resource allocation to unprofitable segments.

Question 3:

What role do demographic variables play here? Follow the instructions in a and b when it comes to demographic variables. How does including demographics affect your conclusion on the profitability difference between customers who used online banking and those not?

Demographics and Profitability Differences Between Online and Offline Customers

Several regression models were estimated to assess the relationship between online banking usage and profitability, controlling for demographic factors.

1. **Model 1 (No Controls)** – The difference between online and offline customers is not statistically significant ($p = 0.21$). Online banking usage alone does not explain any significant variation in customer profitability. So, there is no strong evidence that using online banking increases profitability when no other variables are considered.
2. **Model 2 (With Age, Tenure, and District Controls)** – After controlling for these variables, online customers appear significantly more profitable (~\$20 more on average, $p < 0.001$). Suggests that once you account for some demographic effects (like age, plus missing-age patterns), the online effect grows and becomes significant. AgeMiss has a strongly negative (-36.5) correlation meaning customers with missing age data tend to be significantly less profitable. Each additional year of age, adds about \$1.50 to profit, meaning older customers tend to be more profitable, controlling for other factors.
3. **Model 3 (With Income Instead of Age)** – As we control for income (and district, tenure, etc.), the difference in profitability between online vs. offline shrinks to about 5.01 and is not statistically significant. This suggests that much of the raw gap in profitability was really due to income. The 'IncMiss' variables have a strongly negative (-37.49) impact on profitability. So customers with missing income data tend to be much less profitable. A higher IncMean (i.e., being in a higher income bracket) strongly correlates with greater profitability, even though the per-dollar effect looks small on paper. Over the wide range of income (e.g., from \$12,500 up to \$150,000), it can add up substantially.
4. **Model 4 (Full Controls: Age, Income, Tenure, and District)** – This model provides the best explanation (R-squared: 6.6% -slightly higher than previous models) of profitability differences (adding both age and income adds explanatory power). The

difference between online and offline customers remains positive (~\$13 higher for online users) and statistically significant ($p < 0.01$). This suggests that while income and age explain much of the difference, online banking still contributes positively to customer profitability. Also tenure has a positive and strongly significant (4.73) contribution to profit (Each additional unit of tenure (e.g., each year) adds about 4.74 in profit), longer-tenured customers appear more valuable. And Higher Income Mean and missing income is also highly significant(as explained in model 3)

Model: OLS Adj. R-squared: 0.066
Method: Least Squares F-statistic: 279.2
Date: Thu, 06 Mar 2025 Prob (F-statistic): 0.00
Time: 10:12:32 Log-Likelihood: -2.2124e+05
No. Observations: 31634 AIC: 4.425e+05
Df Residuals: 31625 BIC: 4.426e+05
Df Model: 8
Covariance Type: nonrobust

	coef	std err	t	P> t	[0.025	0.975

[						
--						
Intercept	-80.2973	7.068	-11.361	0.000	-94.150	-66.44
C(District99) [T.1200]	19.5912	5.074	3.861	0.000	9.646	29.53
C(District99) [T.1300]	8.2779	6.244	1.326	0.185	-3.961	20.51
Online99	13.2206	4.600	2.874	0.004	4.204	22.23
AgeMiss	-5.1482	8.183	-0.629	0.529	-21.187	10.89
AgeMean	1.6568	0.115	14.468	0.000	1.432	1.88
IncMiss	-34.2713	8.186	-4.187	0.000	-50.316	-18.22
IncMean	0.0010	4.19e-05	24.951	0.000	0.001	0.00
Tenure99	4.7361	0.191	24.745	0.000	4.361	5.11

Omnibus:	17986.314	Durbin-Watson:	1.996			
Prob(Omnibus):	0.000	Jarque-Bera (JB):	167493.993			
Skew:	2.626	Prob(JB):	0.00			
Kurtosis:	12.974	Cond. No.	5.75e+05			

Question 4:

Based on your analyses, generate conclusions on the profitability difference between customers who used online banking and those not. Does the online channel make customers more profitable? What is your recommendation to the management team?

Conclusions and Recommendations

1. Are online customers more profitable?

Yes, after controlling for all relevant factors, online customers generate approximately \$13 more in profit annually than offline customers. This suggests a real, positive correlation between online banking and higher profitability. The data does not prove

causation, but controlling for demographic variables indicates that online banking likely contributes to increased profitability.

2. Recommendations to Management:

- **Prioritize High-Value Customers** – Pilgrim Bank should focus on high-income, long-tenured, and older customers, as they are the most profitable, aligning with the whale curve in Answer 1, where 21.5% of customers generate 100% of profits. Expanding premium banking services, offering targeted marketing campaigns, and encouraging digital engagement among these segments will enhance profitability. Loyalty rewards and retention strategies should be used to maintain relationships with long-tenured customers, while older customers can benefit from digital literacy programs and personalized financial solutions to further shift more customers into the profitable segment of the curve.
- **Improve Data Collection and Address Missing Income Risks** – Customers with missing income data are \$34.27 less profitable (as indicated by our analysis in Answer 3), making it crucial to enhance data collection efforts. Encouraging income disclosure through targeted incentives, such as better credit offers or personalized banking solutions, can help mitigate profitability risks. Ensuring accurate data at account opening and periodic income verification will strengthen customer segmentation and financial planning.
- **Maximize Digital Banking Engagement While Reducing Costs** – Online banking users are more profitable, making it essential to promote digital adoption among high-value customers. The bank should develop strategies to transition profitable offline users to online banking while limiting high-cost services (access to tellers / ATMs) for persistently unprofitable customers. Shifting low-value customers toward digital self-service options can optimize resources and enhance long-term profitability.

Question 5:

How well does customer channel use in 1999 predict how profitable customers will be in 2000 after controlling for all the other variables?

Impact of 1999 Online Banking Use on 2000 Profitability and Retention

- The regression model shows that Online99 has a coefficient of ~27.90 ($p < 0.001$), meaning that even after controlling for age, income, tenure, and district, online banking use in 1999 is a strong predictor of profitability in 2000.
- While the R^2 is modest (3.7%), the significant coefficient suggests that prior online usage is positively correlated with future profitability.
- On average, being an online-banking customer in 1999 predicted higher 2000 profitability, by roughly \$28, compared to an otherwise similar offline customer. This supports the idea that promoting online banking can boost or sustain customer profitability.

OLS Regression Results

Dep. Variable:	Profit00	R-squared:	0.037
Model:	OLS	Adj. R-squared:	0.037
Method:	Least Squares	F-statistic:	126.7
Date:	Mon, 10 Mar 2025	Prob (F-statistic):	1.72e-209
Time:	01:14:27	Log-Likelihood:	-1.9444e+05
No. Observations:	26396	AIC:	3.889e+05
Df Residuals:	26387	BIC:	3.890e+05
Df Model:	8		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
Intercept	-65.1350	11.038	-5.901	0.000	-86.770	-43.500
C(District99) [T.1200]	22.3846	8.153	2.746	0.006	6.404	38.365
C(District99) [T.1300]	13.8587	10.016	1.384	0.166	-5.774	33.491
Online99	27.8968	7.244	3.851	0.000	13.699	42.095
AgeMiss	2.3779	12.848	0.185	0.853	-22.804	27.560
AgeMean	1.3289	0.175	7.576	0.000	0.985	1.673
IncMiss	-31.2429	12.862	-2.429	0.015	-56.452	-6.034
IncMean	0.0013	6.34e-05	20.763	0.000	0.001	0.001
Tenure99	4.7469	0.300	15.813	0.000	4.159	5.335

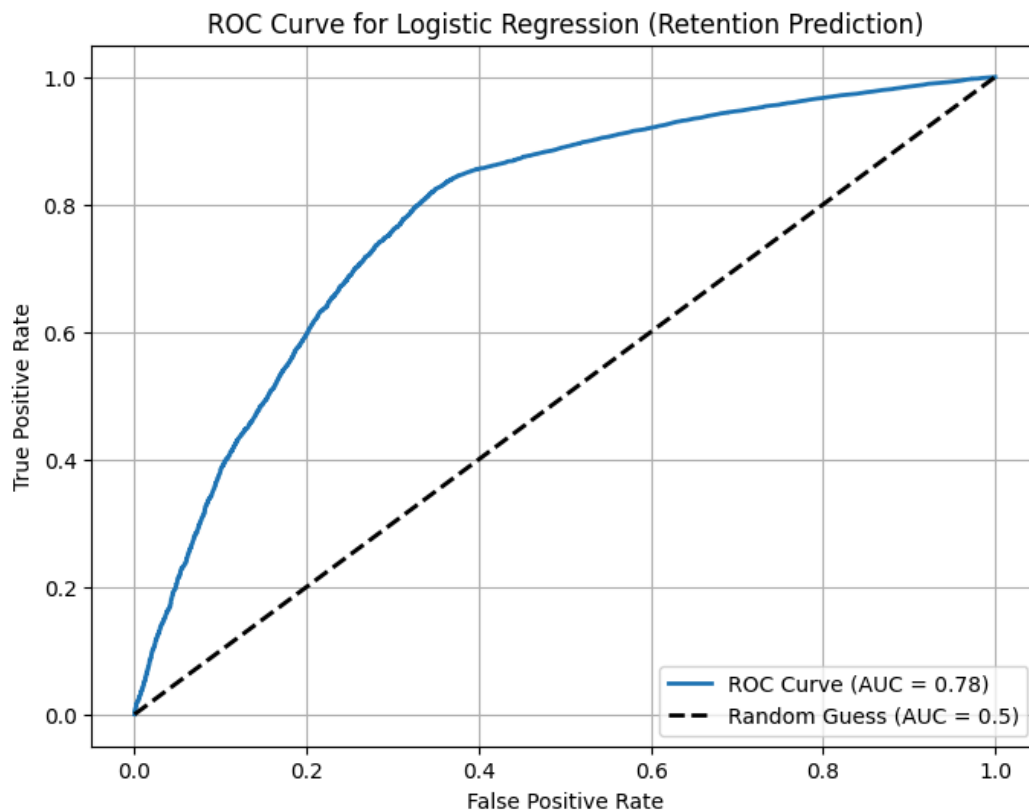
Omnibus:	56410.238	Durbin-Watson:	1.990
Prob(Omnibus):	0.000	Jarque-Bera (JB):	1163256320.930
Skew:	18.238	Prob(JB):	0.00
Kurtosis:	1030.782	Cond. No.	5.74e+05

Question 6:

How well does customer channel use in 1999 predict whether customers stay with the bank through 2000 after controlling for all the other variables?

How well does 1999 channel use predict 2000 retention?

- A logistic regression model estimates the impact of online banking on customer retention. As shown in the ROC curve below, the model achieves an AUC value of 0.78, indicating that it performs significantly better than random guessing. While the model demonstrates good predictive power, it is not perfect, leaving room for improvement.



- Online99 has a coefficient of 0.108, translating to an 11% ($\exp(0.108) \approx 1.11$) higher odds of retention for online users compared to offline users ($p = 0.043$).
- While not a massive effect, it indicates that customers who used online banking in 1999 were more likely to stay with the bank in 2000.
- From a business strategy perspective, this implies that promoting online adoption not only improves profitability but also reduces churn.

Logit Regression Results

Dep. Variable:	Stay00	No. Observations:	31634
Model:	Logit	Df Residuals:	31625
Method:	MLE	Df Model:	8
Date:	Mon, 10 Mar 2025	Pseudo R-squ.:	0.1791
Time:	01:14:28	Log-Likelihood:	-11655.
converged:	True	LL-Null:	-14198.
Covariance Type:	nonrobust	LLR p-value:	0.000

	coef	std err	z	P> z	[0.025	0.975]
Intercept	1.5807	0.088	17.970	0.000	1.408	1.753
C(District99) [T.1200]	0.1052	0.055	1.916	0.055	-0.002	0.213
C(District99) [T.1300]	0.0360	0.068	0.529	0.597	-0.097	0.169
Online99	0.1079	0.053	2.025	0.043	0.003	0.212
AgeMiss	-1.1215	0.076	-14.666	0.000	-1.271	-0.972
AgeMean	0.0071	0.002	4.436	0.000	0.004	0.010
IncMiss	-1.0642	0.076	-13.963	0.000	-1.214	-0.915
IncMean	3.101e-06	6.28e-07	4.938	0.000	1.87e-06	4.33e-06
Tenure99	0.0386	0.002	16.196	0.000	0.034	0.043

Final Takeaways for Pilgrim Bank

- **Profitability is highly skewed**, with a small percentage of customers generating most of the profits.
- **Focusing on profitable customers** could enhance financial performance while reducing reliance on unprofitable segments.
- **Online banking has a positive impact on customer profitability**, though much of the effect is explained by demographics.
- **Encouraging online banking adoption** can lead to **both higher profitability and improved customer retention**.
- **Strategic recommendations** include targeted **digital adoption campaigns**, **removal of barriers to online banking**, and **incentives** to retain online banking customers.